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Advancing EEG-based assessment of consciousness and cognition in prolonged disorders of consciousness

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ABSTRACT

N=44 participants: UWS n=14, MCS n=17, LIS n=11 +2 able-bodied benchmarks (NCT03827187). Structured multi-phase motor imagery BCI with wearable EEG: sensorimotor rhythm modulation with/without neurofeedback + binary Q&A. CRS-R and WHIM behavioral at each session. Results: 73.8% achieve significant decoding accuracy (DA); 90% of those progress to Q&A frequently >70% usability threshold. LIS > MCS (p=0.007) and UWS (p=0.048) on significant runs; UWS > MCS during Q&A (p=0.049) driven by familiar voice. LOSO CV: combined DA+behavioral balanced accuracy 55% -> 62%, MCS sensitivity 39%->69%, LIS sensitivity 78%->67%. Sensorimotor/parietal activity differentiates groups. Potential movement-independent diagnostic tool.

What was actually measured (numbers):

73.8% significant DA, 90% progression, 70% threshold, n=44, balanced accuracy +7%, p values as above

Devices / Protocol:

Wearable EEG (low-cost), motor imagery BCI, CRS-R, WHIM

Replication across teams / Independence:

ClinicalTrials.gov registered, leave-one-subject-out cross-validation, multi-center (Sheffield/Bath)

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